

EdCube Development Plans

Roadmap:

v5.0: Core Infrastructure & Compliance

- ☐ **Implement DMCA Policy:** Establish a formal legal landing page and footer to mitigate takedown risks.
- ☐ **Finalize Dynamic Game Loading:** Migrate from static pages to a single-template dynamic loader for the entire library.
- ☐ **Complete Game Library & Categorization:** Populate the full database of titles and organize them into searchable genres.
 - ☐ Separate all Fireboy & Watergirl games
 - ☐ Fix hollow knight
 - ☐ Fix UBG redirects on tag
 - ☐ Add about this game section on game page (using game description attribute)
- ☐ **Survey System Integration:** Build database and Sheets/Excel integration for user feedback (1-10 scale) with automated statistical analysis (median, average, typical).
- ☐ **Consistency Audit:** Ensure all game page titles, metadata, and site-wide UI elements are accurate and uniform.
- ☐ **Global "Kill Switch":** Implement a centralized mechanism to push maintenance modes or emergency updates across all mirrors.

v5.1: Search, Identity & Stability

- ☐ **Advanced Search Filters:** Add robust filtering options including category, popularity, and rating.
- ☐ **Customization Engine:** Deploy personalization features (Dark/Light mode, custom fonts, various backgrounds, custom cursors/animations).
- ☐ **Primary Infrastructure Migration:** Secure a dedicated hosting provider and a professional Top-Level Domain (TLD).
- ☐ **Auth System v1:** Implement a sign-up/login system featuring 2FA, OAuth, and salted/hashed encryption.
- ☒ **Development Workflow:** Transition to a GitHub Organization with formalized Git version control, PR management, and branch merging.
- ☐ **Automated Backups:** Set up automated cron jobs for SQL database backups to ensure data persistence for the 15k+ user base.

v5.2: Performance & Security

- ☐ **CDN & Scalability:** Integrate Cloudflare for DDoS protection and conduct horizontal vs. vertical scaling tests.
- ☐ **Optimization Suite:** Implement high image compression, lazy loading, caching, and redundant code reduction.
- ☐ **Automated Security:** Deploy automated bundling/minification and integrate GitGuardian CLI for secrets/API key encryption.
- ☐ **Internal Tooling:** Add built-in developer tools (e.g., Eruda) with encrypted access.
- ☐ **Organic Discovery:** Execute in-depth SEO strategies and advanced analytics to drive search-based traffic.
- ☐ **A/B Testing Framework:** Integrate a system to test UI layouts and feature variations against user retention metrics.

v5.3+: Engagement & Expansion

- ☐ **Proxy Integration:** Deploy Ultraviolet or similar proxy tech for bypassing restricted network environments.
 - ☐ **Gamification System:** Add user tokens, a cosmetic shop, daily streaks, and achievement badges.
 - ☐ **Community Features:** Launch user-generated content (reviews, likes/dislikes)
 - ☐ **Cross-Platform Integration:** Develop Google Chat and Discord bots/plugins with full webhook support.
 - ☐ **Social Overhaul:** Revamp the Discord server with tighter website-to-Discord data synchronization.
 - ☐ **AI Implementation:** Integrate an AI chatbot for user support and site navigation.
 - ☐ **Domain Proliferation:** Deploy multiple subdomains and mirrors with streamlined, unified analytics.
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Infrastructure:

1. Hosting & Infrastructure (Cloudflare Ecosystem)

- **Frontend: Cloudflare Pages** (Free) — Optimized Next.js hosting with global edge delivery.
- **Backend: Cloudflare Workers** (Free) — Serverless logic executing at the edge for sub-10ms response times.
- **Storage: Cloudflare R2** (Free) — S3-compatible object storage for game assets and source files with **\$0 egress fees**.

- **Network: Custom Domain & Email** — Wholesale domain registration with free intelligent email routing to Gmail.
- **Acceleration: Cloudflare Hyperdrive** (Free) — Essential middle-layer that pools database connections and caches regional Supabase queries to prevent serverless "cold start" lag.

2. Data Layer (Supabase)

- **Database: Supabase / PostgreSQL** (Free) — Relational data handling for user profiles, game states, and complex queries.
 - **Integration:** Connected via **Hyperdrive** to transform a regional Postgres instance into a globally-distributed data source.
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Features:

- Home page with header, hero banner, featured games carousel, game library, footer
- Settings tab for account and profile settings and customization features
- Each game should be categorized (eg shooter, parkour, etc)
- Game library should have a search feature and filters for sorting (eg by category, name, popularity, etc)
- One game page that dynamically loads the selected game in an iframe with an options bar with likes/dislikes, share, report bug, fullscreen as well as a recommended games section and then comments (with optional ratings) below
- All games should have data associated with them (eg name, id, description, category, popularity, likes, dislikes, etc)
- Login/signup system
- Proxy page using something like Ultraviolet or Scramjet to bypass web filters and browse anonymously
- Chat page where users of EdCube can talk to each other (if logged in then using that account, if not then anonymous)
- Game request page